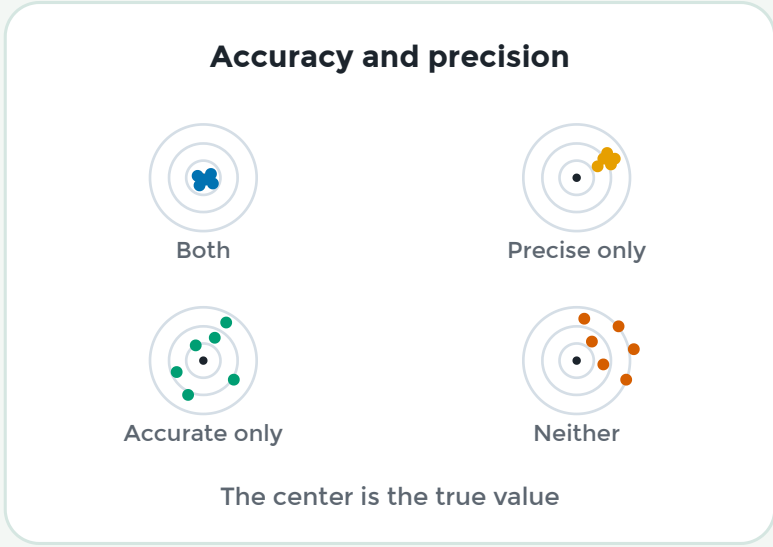
DATA 110 Week 11

Getting to know your data

Look before you trust the chart

1,450 births in North Carolina, 2001



Prof. Rebin Muhammad

Montgomery College

TODAY

Look before you trust a chart

- Can this file answer the question, and where did it come from?
- Is it documented, and can a program read it?
- Find the codes that are not real measurements.
- Then describe it: counts, a summary, and a chart.

SEVEN CHECKS

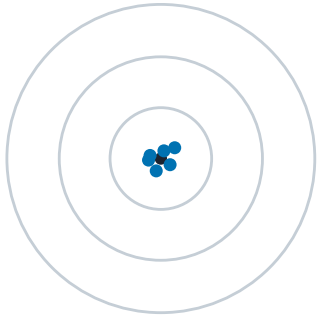
A file can fail in more than one way

Seven questions to ask of a file

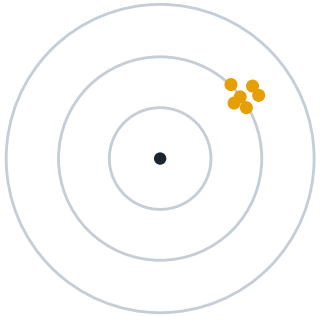
	Ask
Accuracy	Is there a systematic error?
Precision	How much random noise is there?
Usability	Is the file documented and machine-readable?
Completeness	Are rows or values missing?
Consistency	Do related columns agree?
Accessibility	Can you obtain the file?
Relevance	Can it answer your question?

Close to the truth, and close to each other

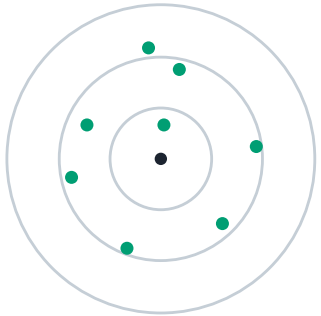
Accurate and precise



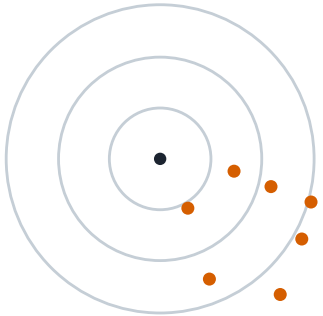
Precise, not accurate



Accurate, not precise



Neither



QUESTION

The shots form a tight cluster, far from the center. What failed?

The shots agree with one another. They do not land on the true value.

THE ANSWER

Accuracy failed. Precision did not.

- Precision is how close the values are to each other.
- Accuracy is how close they are to the true value.
- A measurement can be precise and still be wrong.

USABILITY

A code is unreadable without a list

- A 1 in one column is not a 1 in another column.
- A data dictionary says what each column means, and what each code means.
- The penguins table has a reference page. This birth file needs the same kind of list.

THE FILE

1,450 births in North Carolina

NCbirths.csv. One row is one birth.

THE FILE

Where the rows come from

A sample of births in North Carolina in 2001, selected by John Holcomb from the North Carolina State Center for Health and Environmental Statistics. The file has 1,450 rows and 15 columns.

In this copy, a missing weight gain is written as the text #N/A, and one missing gestation is written as −999.

The columns this lecture uses

Column	What it holds
Weeks	Completed weeks of gestation
Gained	Pounds gained during pregnancy
Sex	1 = male, 2 = female
Smoke	1 = smoker, 0 = not
Plural	1 = single, 2 = twins, 3 = triplets
BirthWeightGm	Birth weight in grams
RaceMom	A number code for the mother's race
MomRace	A word: white, black, hispanic, or other

QUESTION

Gained is blank 40 times. Weeks contains −999 once. Which value changes the mean?

A blank is not a number. −999 is a number.

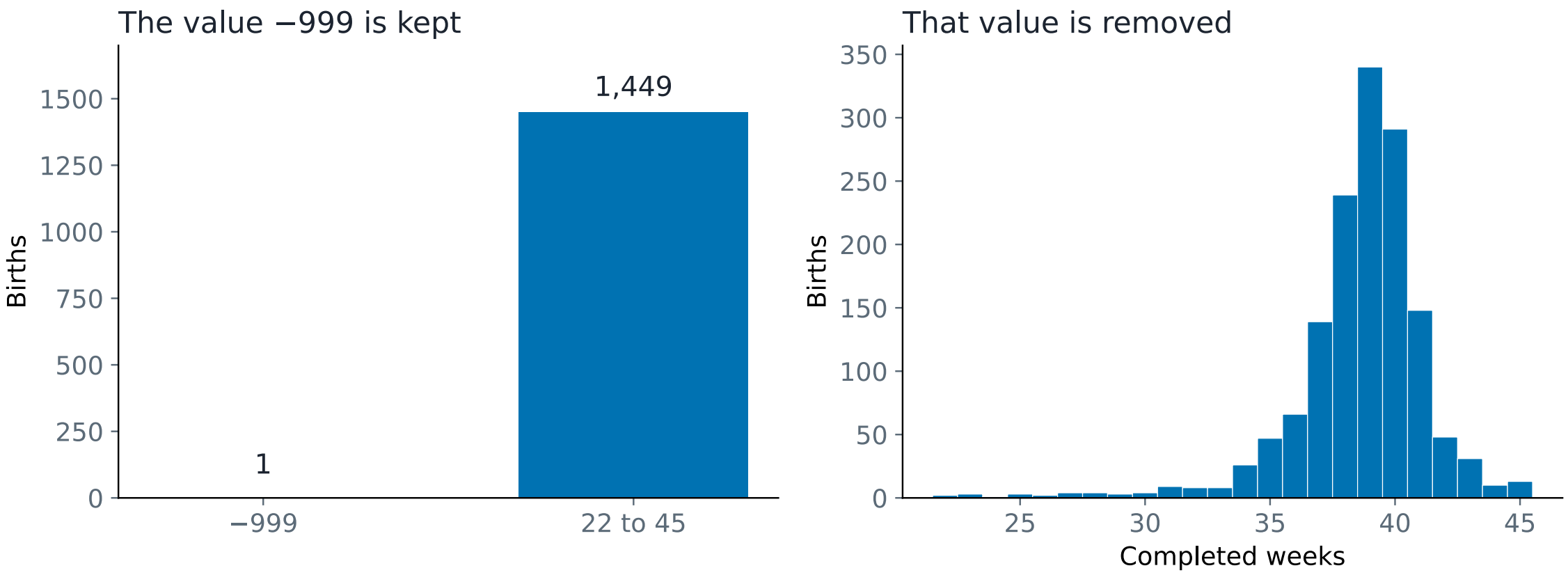
THE MEAN

−999 is included. A blank is not.

- With −999 kept, the mean of weeks is 37.9 and the standard deviation is 27.4.
- With that one value removed, the mean is 38.6 and the standard deviation is 2.7.
- The median stays 39 either way.

One extreme number can dominate a mean and a standard deviation. It does not have to move the median.

One number hides the rest



340 of the 1,449 real records are exactly 39 weeks. Completed weeks in this file run from 22 to 45.

CONSISTENCY

Two columns, one idea

RaceMom is a number. MomRace is a word.

TWO DESCRIPTIONS

Code 5 does not match the other column

- The column notes say code 5 means Japanese.
- All 164 rows with code 5 have MomRace equal to hispanic.
- On those rows, HispMom is Mexican (128), Central or South American (23), Puerto Rican (8), other Hispanic (3), or Cuban (2).

None of the 164 are marked “not Hispanic.”

QUESTION

The notes say Japanese. MomRace says hispanic. What do you do?

Code 5 has two descriptions, and they disagree.

THE CHECK

You stop. You do not pick one in silence.

- A dictionary can disagree with another column in the same file.
- HispMom agrees with MomRace on these 164 rows. It does not agree with the word Japanese.
- Record the contradiction. Recode a code only after the descriptions agree.

CODES THAT AGREE

Sex, smoking, and single or twins

- Sex: 1 is male, 744 births. 2 is female, 706 births.
- Smoke: 0 is not a smoker, 1,236. 1 is a smoker, 209. The value is missing 5 times.
- Plural: 1 is a single birth, 1,401. 2 is twins, 45. 3 is triplets, 4.

These codes match the column notes. Sex and Plural add to 1,450. Smoke adds to 1,445, plus the 5 blanks.

map replaces each code with a word

```
codes = {1: "male", 2: "female"}
births["sex_word"] = births["Sex"].map(codes)
```

map looks up each number in the dictionary. A missing value has nothing to look up, so it stays missing.

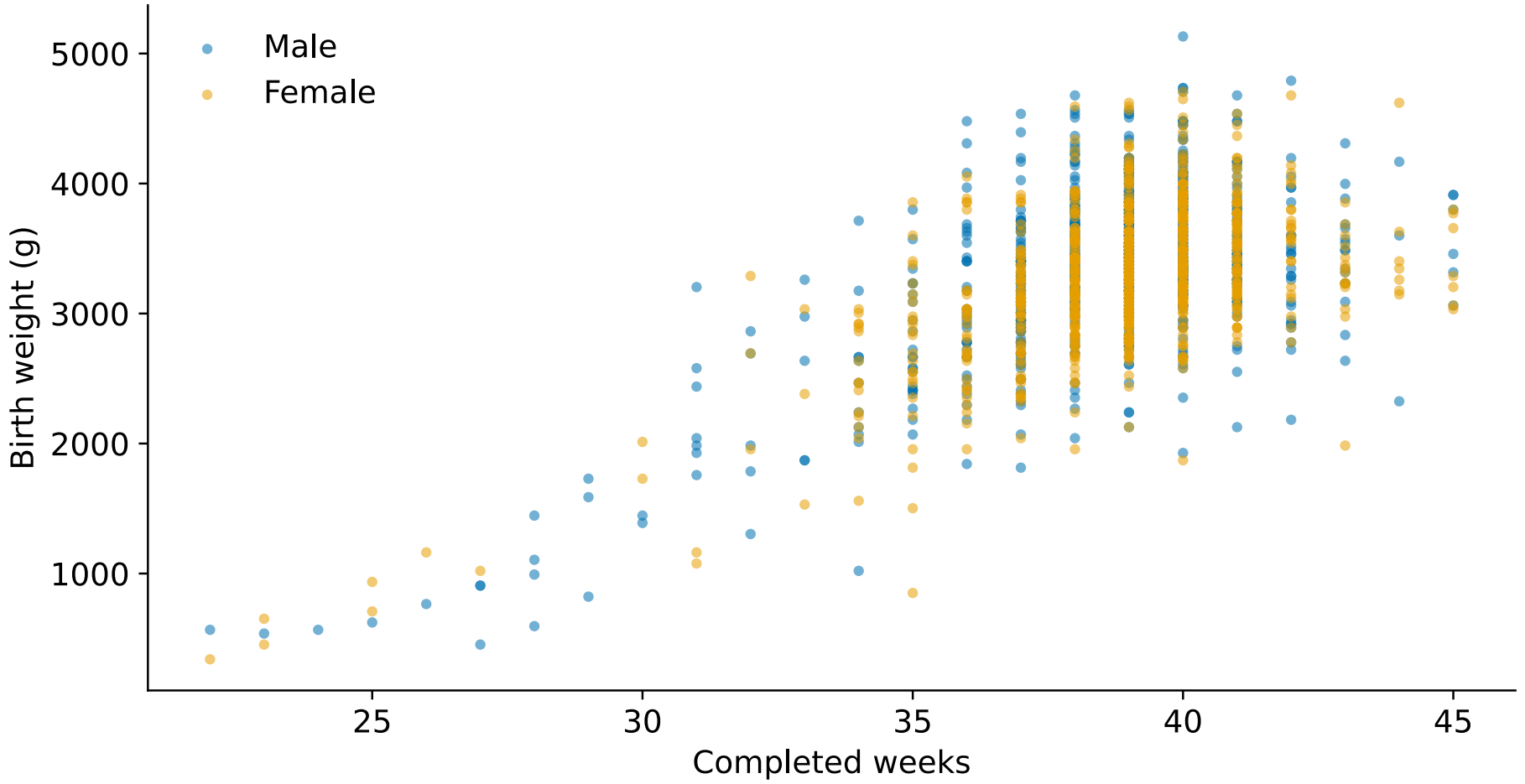
THE SENTINEL

A negative week is not a gestation

```
births = births.copy()
births.loc[births["Weeks"] < 0, "Weeks"] = pd.NA
```

copy keeps the original table. loc selects the one negative row and writes a blank in Weeks.

Weight against a real gestation



WHAT AGREES

Two rules hold on the real numbers

- Low is 1 exactly when the weight is 2,500 g or less. That is true on all 1,450 rows. 125 births are at or under the line.
- Premie is 1 exactly when weeks are 36 or less. That is true on all 1,449 rows with a real week. 190 births are at or under 36 weeks.
- The row with −999 is also flagged premature. That flag cannot be checked against a real week.

FURTHER READING

The list, and the blanks

- A data dictionary: what a column and a code mean
- pandas: missing values are skipped by mean, unless the blank was stored as a number
- The penguins reference page, a dictionary the class already uses

Figures for Data 110, Fall 2026. Run make_figures.py to rebuild every figure.